Exam 1 Solutions - June 2026

BUS 411: Fixed Income Securities

A. Short Answer Questions

Question A1

A pension fund is choosing between a newly issued 8-year fixed-rate corporate bond, an 8-year floating-rate note indexed to SOFR, and an 8-year callable agency bond.

Explain how the main risk exposures differ across the three securities. Your answer should address interest-rate risk, reinvestment risk, credit or spread risk, and embedded-option risk. Also explain why a security with more stable price behavior can still be risky for the investor.

Solution.

The fixed-rate corporate bond has predictable contractual coupons and principal, but its price is sensitive to changes in interest rates and credit spreads. It also has credit risk because the issuer may deteriorate or default.

The floating-rate note has lower price duration because its coupon resets with SOFR. That does not make it risk-free: the coupon can fall when short rates fall, the issuer can still default, liquidity can weaken, and any cap or floor changes the investor's exposure.

The callable agency bond has interest-rate risk and embedded-option risk. When rates fall, the issuer is more likely to call the bond, which limits price appreciation and forces reinvestment at lower yields. When rates rise, the call becomes less relevant and the bond behaves more like a longer-duration security.

Stable price behavior is not the same as low investor risk. A floating-rate note may trade near par after reset but still expose the investor to lower future income, spread widening, credit deterioration, and liquidity risk.

Question A2

Explain why yield to maturity is not the same thing as the realized return earned by a bond investor. In your answer, identify at least three assumptions embedded in yield to maturity and give one example where a bond with a higher yield to maturity may produce a lower realized return than another bond.

Solution.

Yield to maturity is the single discount rate that equates promised cash flows to price. It is not a guaranteed realized return.

YTM assumes the investor holds the bond to maturity, all promised cash flows are paid on schedule, coupons are reinvested at the YTM, and the yield curve is summarized by one constant rate. For callable or prepayable bonds, it also ignores that cash-flow timing may change.

A higher-YTM callable bond can produce a lower realized return if rates fall and the issuer calls the bond. The investor receives the call price, loses future high coupons, and reinvests at lower market yields. A lower-YTM noncallable bond may then outperform over the same horizon.

Question A3

A manager says: "My portfolio has a duration of 5.4, so I understand its interest-rate risk."

Critique this statement. Explain what duration captures, what it misses, and why convexity and key rate duration can change the risk assessment. Use the intuition of a nonparallel yield-curve shift in your answer.

Solution.

Duration measures the approximate percentage price change for a small parallel yield change. A duration of 5.4 means a 100 bp parallel increase in yields implies roughly a 5.4% price decline before convexity adjustment.

The statement is incomplete because duration misses curvature and yield-curve shape risk. Convexity matters for larger rate moves because price-yield relationships are nonlinear. Key rate duration matters because yield curves usually move nonparallel. For example, if the portfolio has large 10-year key rate exposure but the benchmark has more 2-year exposure, both may have similar total duration while reacting very differently to a bear steepener or flattening move.

Duration is a useful first-order measure, not a full description of interest-rate risk.

Question A4

Investment-grade corporate bonds usually trade at positive spreads over Treasury securities with similar maturity.

Explain the economic reasons for this spread. Your answer should distinguish compensation for expected default loss, downgrade risk, liquidity risk, taxes or regulation when relevant, and market risk premia. Then explain why the observed spread is not a pure forecast of default losses.

Solution.

Corporate spreads compensate investors for several risks beyond Treasury rates:

- Expected default loss: probability of default times loss given default.
- Downgrade risk: price decline if credit quality worsens before default.
- Liquidity risk: compensation for wider bid-ask spreads and difficult trading, especially in stress.

- Tax, regulatory, and balance-sheet effects: different investors face different after-tax values and capital charges.
- Risk premia: compensation for bearing systematic credit risk that tends to perform poorly in recessions.

The observed spread is not a pure default forecast because it contains liquidity premia, risk premia, supply-demand effects, covenant and seniority differences, and compensation for mark-to-market volatility.

Question A5

A bond portfolio manager is benchmarked to a broad bond index with thousands of securities. The manager is considering either a cell-based construction approach or a multi-factor model approach.

Compare the two approaches. Explain how each approach controls benchmark-relative risk, why pure bond index replication is difficult, and why tracking error rather than total portfolio volatility is the central risk measure for this manager.

Solution.

The cell-based approach divides the benchmark into buckets such as duration, maturity, sector, credit quality, coupon, callability, and mortgage characteristics. The portfolio is built to match benchmark weights across those cells. It is intuitive and easy to explain, but it can miss correlations across cells and can create unintended factor bets.

A multi-factor model represents bond returns through common drivers such as Treasury curve shifts, spread changes, volatility, mortgage prepayment factors, sector factors, and issuer effects. It measures how the portfolio differs from the benchmark in factor exposure and estimates forward-looking tracking error.

Pure replication is difficult because broad bond indexes contain thousands of securities, many bonds are illiquid, index prices may not be executable, bid-ask spreads create drag, and cash flows must be reinvested. Therefore, managers usually sample the index or use enhanced indexing.

Tracking error is central because the manager is judged relative to a benchmark. Total volatility can be high for both portfolio and benchmark while active risk is low. The relevant question is how much the portfolio can deviate from benchmark returns.

B. Problems

Problem B1

A 6-year corporate bond has face value \$1,000, a 5.40% annual coupon paid semiannually, and a yield to maturity of 5.90% with semiannual compounding. Settlement occurs 74 days after the last coupon date in a 182-day coupon period. The matched 6-year Treasury yield is 4.35%.

Answer the following:

a. Compute the clean price of the bond per \$1,000 face value.

b. Compute accrued interest and the full price.

c. Compute the bond's yield spread over the matched Treasury yield.

d. Suppose the bond has modified duration of 5.05 and convexity of 31.8. Approximate the percentage price change if the bond's yield rises by 75 basis points.

e. Explain why the duration-convexity approximation may be less reliable for this corporate bond than for an option-free Treasury bond.

Solution.

The semiannual coupon is:

$$C = \frac{0.054}{2} \times 1000 = 27.$$

The semiannual yield is:

$$y = \frac{0.059}{2} = 0.0295.$$

There are 12 remaining semiannual periods. The clean price is:

$$P = 27\left[\frac{1-(1.0295)^{-12}}{0.0295}\right] + \frac{1000}{(1.0295)^{12}} = 975.04.$$

Accrued interest is:

$$AI = 27 \times \frac{74}{182} = 10.98.$$

The full price is:

$$975.04 + 10.98 = 986.02.$$

The yield spread over the matched Treasury is:

$$5.90\% - 4.35\% = 1.55\% = 155 \text{ bps}.$$

Using duration and convexity:

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\Delta y + \frac{1}{2}C(\Delta y)^2.$$

With $\Delta y = 0.0075$:

$$\frac{\Delta P}{P} \approx -5.05(0.0075) + \frac{1}{2}(31.8)(0.0075)^2 = -0.03698.$$

The approximate price change is -3.70%, so the clean price is approximately:

$$975.04(1 - 0.03698) = 938.98.$$

The approximation may be less reliable than for an option-free Treasury because corporate prices can move from both Treasury-rate changes and spread changes. Liquidity, downgrade risk, and changing credit risk can alter the spread at the same time yields move.

Problem B2

The following annual-pay Treasury securities are observed in the market. Assume all rates are annual effective rates and face value is $100.

Maturity	Coupon rate	Price
1 year	0.00%	96.1538
2 years	5.00%	100.20
3 years	6.00%	101.10

Answer the following:

a. Compute the 1-year spot rate.

b. Bootstrap the 2-year spot rate.

c. Bootstrap the 3-year spot rate.

d. Compute the 1-year forward rate from year 2 to year 3.

e. Use the spot curve plus a constant 120 basis point spread to price a 3-year annual-pay corporate bond with a 5.50% coupon and face value $100. Briefly explain what the constant spread assumption means.

Solution.

For the 1-year zero:

$$96.1538 = \frac{100}{1 + z_1},$$

so:

$$z_1 = 4.00\%.$$

For the 2-year 5% coupon Treasury:

$$100.20 = \frac{5}{1.04} + \frac{105}{(1+z_2)^2}.$$

Solving:

$$z_2 = 4.9151\%.$$

For the 3-year 6% coupon Treasury:

$$101.10 = \frac{6}{1.04} + \frac{6}{(1.049151)^2} + \frac{106}{(1+z_3)^3}.$$

Solving:

$$z_3 = 5.6529\%.$$

The 1-year forward rate from year 2 to year 3 is:

$$f_{2,3} = \frac{(1+z_3)^3}{(1+z_2)^2} - 1.$$

Thus:

$$f_{2,3} = \frac{(1.056529)^3}{(1.049151)^2} - 1 = 7.1440\%.$$

With a constant 120 bp spread, the discount rates are:

$$5.20\%, \quad 6.1151\%, \quad 6.8529\%.$$

The corporate bond price is:

$$P = \frac{5.5}{1.052} + \frac{5.5}{(1.061151)^2} + \frac{105.5}{(1.068529)^3} = 96.59.$$

The constant spread assumption means every maturity point on the Treasury spot curve is shifted up by the same number of basis points. It is a simple credit-spread valuation method, not a full model of default timing, liquidity, or curve-specific spread risk.

Problem B3

A bond portfolio is managed relative to a benchmark. A factor model reports the following active exposures and monthly factor volatilities. Assume the three factors are uncorrelated.

Risk factor	Active exposure	Monthly factor volatility
Treasury curve level	0.25	24 bps
Corporate spread	0.50	18 bps
Securitized spread	-0.15	20 bps

Idiosyncratic tracking error is 6.0 bps per month. The client's monthly tracking-error budget is 14.0 bps.

The manager can reduce the corporate spread active exposure from 0.50 to 0.25 by buying more benchmark-like corporate bonds. This trade costs 2.0 bps of portfolio value and leaves the other exposures and idiosyncratic tracking error unchanged.

Answer the following:

a. Compute the isolated monthly tracking error contribution for each factor.

b. Compute total systematic tracking error.

c. Compute total monthly tracking error after adding idiosyncratic tracking error. Is the portfolio inside the risk budget?

d. If Treasury yields rise by 30 bps, corporate spreads widen by 20 bps, and securitized spreads tighten by 10 bps, estimate the active return impact in basis points.

e. Compute total monthly tracking error after reducing the corporate spread active exposure to 0.25. Explain the trade-off between the tracking-error reduction and the 2.0 bps trading cost.

Solution.

The isolated tracking error values are the absolute value of active exposure times factor volatility:

Risk factor	Calculation	Isolated TE
Treasury curve level	0.25×24	6.00 bps
Corporate spread	0.50×18	9.00 bps
Securitized spread	$\lvert -0.15 \rvert \times 20$	3.00 bps

Because the factors are assumed uncorrelated, systematic tracking error is the square root of the sum of squared isolated tracking errors:

$$TE_{\text{sys}} = \sqrt{6.00^2 + 9.00^2 + 3.00^2} = \sqrt{126} = 11.22 \text{ bps per month.}$$

Total tracking error adds idiosyncratic tracking error in variance terms:

$$TE_{\text{total}} = \sqrt{11.22^2 + 6.00^2} = 12.73 \text{ bps per month.}$$

The portfolio is inside the 14.0 bps monthly tracking-error budget.

The active return impact is approximated as:

$$\Delta R_{\text{active}} \approx -(\text{active exposure})(\text{factor shock}).$$

Factor	Calculation	Active return impact
Treasury curve level	$-0.25(30)$	-7.50 bps
Corporate spread	$-0.50(20)$	-10.00 bps
Securitized spread	$-(-0.15)(-10)$	-1.50 bps

Total estimated active return impact is:

$$-7.50 - 10.00 - 1.50 = -19.00 \text{ bps.}$$

After reducing corporate spread active exposure to 0.25, the isolated tracking errors are:

$$\text{Treasury curve} = 6.00, \quad \text{Corporate spread} = 0.25(18) = 4.50, \quad \text{Securitized spread} = 3.00.$$

The new systematic tracking error is:

$$TE_{\text{sys, new}} = \sqrt{6.00^2 + 4.50^2 + 3.00^2} = 8.08 \text{ bps per month.}$$

The new total tracking error is:

$$TE_{\text{total, new}} = \sqrt{8.08^2 + 6.00^2} = 10.06 \text{ bps per month.}$$

The trade reduces total tracking error by:

$$12.73 - 10.06 = 2.67 \text{ bps per month.}$$

The trade costs 2.0 bps. It is reasonable if the manager wants a larger cushion below the risk budget or has lower conviction in the corporate spread overweight. It may not be necessary if the manager is comfortable with the current 12.73 bps tracking error and wants to preserve the active credit view. The trade-off is immediate transaction cost and less active exposure versus lower predicted active risk.

Problem B4

A \$100 million bond portfolio is benchmarked to a broad aggregate index. The current sector weights are:

Sector	Portfolio weight	Benchmark weight
Treasury	28%	35%
Government/agency	7%	10%
Corporate	45%	30%
Securitized	20%	25%

The current portfolio duration is 5.95 versus benchmark duration of 5.50. The current portfolio spread duration is 5.90 versus benchmark spread duration of 5.20. The risk budget allows monthly predicted tracking error of at most 12.0 bps, duration mismatch within +/-0.25 years, spread-duration mismatch within +/-0.50 years, and issuer active weight no greater than 3.0%.

The current predicted tracking error is 16.8 bps per month, and the largest issuer active weight is 4.2%. The manager proposes the following rebalance:

Trade	Market value	Sector	Duration	Spread duration
Sell BBB financial corporate bond	\$8 million	Corporate	6.2	6.0
Buy 2-year Treasury note	\$5 million	Treasury	1.9	0.0
Buy agency MBS pool	\$3 million	Securitized	3.2	3.0

After the rebalance, the risk model estimates predicted tracking error of 11.4 bps per month and largest issuer active weight of 2.8%. Transaction cost is 20 bps of total traded market value.

Answer the following:

a. Compute the active sector weights before the rebalance.

b. Compute the active sector weights after the rebalance.

c. Estimate the portfolio duration and spread duration after the rebalance.

d. Determine whether the rebalance satisfies the stated risk constraints.

e. Compute the transaction cost in dollars and in basis points of portfolio market value. Explain the trade-off the manager is making.

Solution.

Active sector weights before rebalancing are portfolio weight minus benchmark weight:

Sector	Before active weight
Treasury	-7%
Government/agency	-3%
Corporate	+15%
Securitized	-5%

After selling \$8 million corporates, buying \$5 million Treasuries, and buying \$3 million securitized bonds, the new portfolio weights are:

Sector	New portfolio weight	Benchmark weight	After active weight
Treasury	33%	35%	-2%
Government/agency	7%	10%	-3%
Corporate	37%	30%	+7%
Securitized	23%	25%	-2%

The duration change is approximated from the market-value trade weights:

$$\Delta D = -0.08(6.2) + 0.05(1.9) + 0.03(3.2) = -0.305.$$

So the new portfolio duration is:

$$5.95 - 0.305 = 5.645.$$

The new duration mismatch is:

$$5.645 - 5.50 = 0.145,$$

which is within the +/-0.25 limit.

The spread-duration change is:

$$\Delta SD = -0.08(6.0) + 0.05(0.0) + 0.03(3.0) = -0.390.$$

So the new spread duration is:

$$5.90 - 0.390 = 5.510.$$

The spread-duration mismatch is:

$$5.510 - 5.20 = 0.310,$$

which is within the +/-0.50 limit.

The rebalance satisfies the stated constraints:

Constraint	After rebalance	Limit	Satisfied?
Predicted tracking error	11.4 bps/month	<= 12.0	Yes
Duration mismatch	+0.145	+/-0.25	Yes
Spread-duration mismatch	+0.310	+/-0.50	Yes
Largest issuer active weight	2.8%	<= 3.0%	Yes

Total traded market value is:

$$8 + 5 + 3 = 16 \text{ million.}$$

Transaction cost is:

$$0.0020 \times 16{,}000{,}000 = 32{,}000.$$

As a percentage of the $100 million portfolio:

$$\frac{32{,}000}{100{,}000{,}000} = 0.00032 = 3.2 \text{ bps.}$$

The manager pays 3.2 bps in transaction cost to reduce active corporate concentration, bring predicted tracking error inside the risk budget, and correct duration and spread-duration mismatches. The trade-off is lower benchmark-relative risk at the cost of immediate trading drag and possibly giving up some active credit view.